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Start a systemd service inside chroot from a non systemd based rootfs

Asked 8 years, 7 months ago Modified 2 months ago Viewed 106k times

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17

With init scripts (or with openrc) I alway could run services from a different installation root. but when I run `chroot /somepath/to_root /usr/bin/systemctl start someservice` I got:

```
Running in chroot, ignoring request.
```

Is there a way to force systemd run the service?

Update:
I forgot to say my host system run init scripts or openrc, but never systemd, and that I use chroot to troubleshot Unix systems which can't even launch a minimal shell.

unix systemd chroot sysvinit openrc

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edited Mar 7, 2021 at 19:42

asked Dec 15, 2013 at 16:38
user2284570
1,680 7 31 58

1 I also need to run services into a chroot, it always worked before openrc2, seems impossible now ;(– neofutur Aug 3, 2014 at 1:05

You're trying to solve the wrong problem. If you have OpenRC, you need to convert the systemd service into an OpenRC service. There's really no way around that. – Daniel B Mar 10, 2015 at 21:22

@DanielB : NO! Did you ever heard of systemrescuecd? – user2284570 Mar 11, 2015 at 0:19

No. I also don't see how it relates to your question. – Daniel B Mar 11, 2015 at 6:34

8 Answers

Sorted by: Highest score (default)

41

A well-known problem in systemd distros (Arch Linux, OpenSUSE, Fedora).

Systemd replaces sysvinit, and provides one great advantage over this. In sysvinit, when you ask a service to start, it inherits the execution context of the person invoking the script, which includes environment variables, ulimits, and so on. Systemd improves on this at the contrary by notifying a daemon, which will start the service in a well-defined, healthy, constant environment, where of course the performances of the services are much easier to predict, since the environment is always the same.

This implies that, when I call systemctl from within the chroot, it is irrelevant that I am inside chroot, the environment that will be inherited is still that of PID 1, not my current one. But it gets worse than this: since communication sockets are placed inside /run/systemd, a process in a chroot will not even be able to talk to the init system!

So how do you go about chroot'ing in systemd distros?

1. If all you want to do is have a Linux container, this Arch Wiki page will tell you how to set up a Linux container in less than 30 seconds, thanks to `systemd-nspawn`.
2. If instead you really want a chroot environment, this beautiful and crystal clear Web page will provide you with two working solutions (the second one is a modified version of the one offered at point #1).

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edited Jul 21, 2018 at 21:20
Evan Carroll
7,996 17 72 120

answered Dec 15, 2013 at 17:15
MariusMatutiae
45.7k 10 76 127

2 That I never use systemd in my host system root. In my case I can't mix systemd with openrc.
– user2284570 Dec 16, 2013 at 18:05

2 @TwoD That won't work. Running `systemd-nspawn` fails with "Not running on a systemd system." unless the host is using systemd as well. – hvd Jul 15, 2015 at 11:59

1 @TwoD And I responded because it doesn't sound like that at all to me. :) "I can't run it" is an odd thing to say if you're having trouble finding the executable, which is why I suspect that the problem is what I put in my comment: running it gives that error message and doesn't do anything useful. But even if it turns out the problem really was where to find `systemd-nspawn`, then pointing into the new root won't help. Either the host already has it (because it's running systemd), in which case the host version can be used, or the host doesn't have it, but the new root's version won't work. – hvd Jul 15, 2015 at 12:45

1 `systemd` will refuse to be run in `chroot` – Erkin Alp Güney Mar 11, 2017 at 14:30

1 @CraigHicks Except since you set up a jail, you can’t do things like connecting wayland apps to the host’s compositing. Plain chroot isn’t a jail, but a way to change access search paths for the binaires.
– user2284570 Aug 8, 2020 at 10:02

11

Several years later I must admit there is only one solution to most Systemd practical problems. Because the error is Systemd itself

I am really fed up with Systemd as I had problems that I never faced with things like Upstart or Openrc :

- The enforcing of a kernel requiring cgroups support *(instead of being made optional but enabled by default inside a config file)* **even for embedded systems with only 24Mb of ram** and no writable storage.
  While proponents argue that not being able to use it in such case is the wanted behavior since it wasn't designed for such case, **the reality is the other init systems also don't care while in systemd such feature isn't a separate project which highlights once again the consequences of not following kiss**.
- Despite the claim of being modular, at runtime the dependency hell makes it a strong god object : want to boot over a single reiser4 rootfs ? It's not possible because many programs requires `systemd-udevd` which requires `systemd-init` which requires the `systemd-boot` package which can't be installed at the same time than `grub2` nor can read kernel images from a reiser4 partition.
- Want to connect to the internet through Bluetooth dialup ? If it doesn't work with your Samsung java me phone, then you aren't able to run the scripts and command line software that previously worked manually because of `networkd` .
- Though I recognize the biggest problem is if you are building and maintaining your own Linux distribution : the systemd init module itself has so many dependencies that you can't propose to choose another init system through different install packages.
- The default use of Google dns for systemd if no resolver is defined in terms of privacy.
- The inability to launch a background non service process and log out as there are no simple ways to do it in a new systemd scope.
- Good luck for viewing logs if you can't chroot in your system or if you upgraded from libdb4.8 *(whereas at least, in worst case Microsoft has it's log files in xml format)*.

The only solution :

Systemd is unnecessary complex for solving problems : like alsa instead of ossv4. So if you have something that uses systemd just wipe all the data :

```
dd if=/dev/urandom of=/dev/dm-0 bs=1M count=1
```

and install something that doesn't use it at all while solving problems of SysV Init **like Gentoo with Openrc**.

Concerning my question systemd makes things like the Windows® registry : if a part of it gets screwed up, then it's over.

Share Improve this answer Follow edited May 26 at 23:26 community wiki 11 revs, 2 users 76% user2284570

9 **Please recognize that the design of something can really prevent from getting an answer so that the answer is to switch to something which works**. And that this is a real answer. – user2284570 Nov 18, 2018 at 21:47

2 I had the same opinion, now I am on a little bit more balanced view. Systemd has the super-big advantage that *it can really kill what should be killed*. It is because it tracks all the forked sub-processes with the kernel cgroup feature. None of the older tools can do that. Furthermore, do you remember the crap of the scripts in /etc/init/*.sh?I too, but it is only a bad memory to me today. The systemd service files are clear and around 10 lines long *configs*. Not 200 line long *scripts*. These huge advantages has the systemd, I agree that its all other features are disadvantage. – peterh Mar 25, 2019 at 12:53

1 @peterh : unfortunately not everyone share it I mean to the point of deleting post. This is not about SysV init against Systemd but more against things like Openrc or even Upstart (which allows for short startup scripts as well as parallel service start). At least I learned one thing : Darwin is mostly the os of Apple™ Windows is the os of Microsoft and Linux design is mostly run by red hat. Though SysV init while being slower doesn't restrict you at what you can do at runtime. – user2284570 Mar 26, 2019 at 13:08

1 After many years and at least in Debian I've found the easiest solution is to simply, in the chroot, `apt purge systemd` and then `apt install sysvinit-core` . Since in a chroot systemd is not really "running" even if it's started, uninstalling it won't cripple the chroot, and then you can operate with sysv just as you do in the outside machine. (Unfortunately, in Debian I've found no way to debootstrap chroots with sysvinit preinstalled) – Luis Machuca Apr 28, 2020 at 20:05

1 @LuisMachuca which is the problem of this question (but apt-get can work without Systemd). How do you manage to not have `pid 1` based on a systemd binary because of packages dependencies? This is for that impossibility that devuan was started. Are you still running Jessie or older? – user2284570 Apr 28, 2020 at 22:12

10

`systemd` only ignores "services", so I just run the daemon commands manually.

So instead of

```
service sshd start
```

I use

```
/usr/sbin/sshd -D &
```

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edited Nov 28, 2016 at 14:04
DavidPostill ♦
144k 75 336 371

answered Dec 6, 2015 at 10:36
johnP
131 1 3

2 This doesn't work for all services. Some require to be started as part of the system service starter like Xorg. – user2284570 Dec 6, 2015 at 11:15

`startx` will work for `Xorg` . – Erkin Alp Güney Mar 11, 2017 at 14:26

1 @ErkinAlpGüney : not in chroot... Because of Dbus. – user2284570 Apr 20, 2018 at 19:02

3

No. Services are executed by systemd (pid 1), not by systemctl directly (which only sends a start request), and since systemd runs outside the chroot, so will the service.

Although technically it could be possible to implement this (by making systemctl somehow pass its root to systemd), it is somewhat unlikely to happen since there already is a tool for creating full containers (`systemd-nspawn /somepath/to_root`). You could always contact the mailing list though.

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answered Dec 15, 2013 at 17:00
user1686
378k 58 786 862

1 Nice, But I need to use systemctl since My host system use oepnrc. I want full independent solution – user2284570 Dec 15, 2013 at 17:55

3 I'll muddy the waters yet further by saying: Psst! Mention `RootDirectory=` as well since you are so dangerously short of upvotes. (-: – JdeBP Dec 16, 2013 at 19:56

@JdeBP : What is the difference (in term of results) beetween the variable `RootDirectory` and the `chroot` command? – user2284570 Dec 17, 2013 at 21:35

@grawity : So What append if the `pid 1` is init? – user2284570 Dec 21, 2013 at 9:41

2

Faced this problem once tried to bring up network in rescue mode using network configuration from chroot. Finally this works for me:

```
service --skip-redirect <service> restart
```

or:

```
SYSTEMCTL_SKIP_REDIRECT=_ /etc/init.d/<service> restart
```

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answered Dec 12, 2017 at 14:01
reddot
398 3 7

Nice. But it works only with legacy Init compatible services *(won't work for networking in Fedora rawhide)*. As I said in my answer, the real solution is to screw up anything that use systemd.
– user2284570 Dec 12, 2017 at 14:59

1

Define a service file outside chroot that execute the service inside the chroot with the options `RootDirectory=/path/to/chroot` in conjunction with `MountAPIVFS=on`, the link above contains the magic.

Run systemd script in chroot from outside the chroot

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answered Sep 8, 2020 at 7:19
user avata Antonio Orizondo Leyva
11 1

It will then be run with isolation preventing it from interacting with things like udev on the host.
– user2284570 Sep 8, 2020 at 7:38

its almost a nice solution, but sadly too much of a hack to succesfully solve many problems with it.
– Florian Heigl May 26 at 23:27

-2

If you are launching an inetd-style service with socket activation, consider launching stunnel instead with a configuration file that specifies both a chroot and your binary as an inetd-style launch target.

Note that you may have SELINUX issues. On an Oracle Linux 7.1 system, I had to "chcon -v --type=stunnel_etc_t" on all files that stunnel needed to read.

You will need to use TLS encryption on the client side of the socket (i.e., another stunnel with "client=yes" in the config). Let me know if you want more details on this.

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answered Sep 14, 2015 at 17:11
chas
1

no it's about things like d-bus. I do it in order to diagnose problems on the target chroot.
– user2284570 Sep 14, 2015 at 17:15

-2

You can use `nohup` command to start services in chroot. To start `httpd` service for example, I do it like this.

```
nohup httpd /dev/null &
```

to stop it `pkill httpd`

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answered Feb 26, 2016 at 23:24
ellooku
9 5

What about services like Dbus that can only be started by the installed binary systemd script ?
– user2284570 Apr 11, 2016 at 10:01

You can start such services from its directory with start command. – ellooku Apr 21, 2016 at 18:18

Which is a symlink against systemctl. So it doesn't work. – user2284570 Apr 21, 2016 at 18:25

I do this all the time on Fedora running on my Android. May be I don't know what your problem is.
– ellooku Apr 21, 2016 at 18:32

1 The consequence is this message : `Running in chroot, ignoring request.` . I don't think you do it all the time though chroot. Indeed, the startup script requires systemd. – user2284570 Apr 21, 2016 at 18:35